

# An Exact Cooperation Formula for Introspection Dynamics in the Heterogeneous Public Goods Game

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## Abstract

We study introspection dynamics on multiplayer games in which the payoff difference  $\Delta f_i$  evaluated by a player when considering a strategy switch is independent of all other players' current actions, a property we call *state-independence*. Under this condition the introspection Markov chain decomposes as a random-scan product of  $N$  independent two-state chains, one per player, and the stationary distribution is a product measure. As our main application we consider the heterogeneous public goods game, where  $N$  players may differ in their contributions  $\alpha_i$ , public goods multipliers  $r_i$ , and choice intensities  $\beta_i$ . We prove that the linear payoff structure implies state-independence, and the long-run cooperation probability admits the exact closed form

$$p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N \left[ \frac{1 - \mu_{i0} - \mu_{i1}}{1 + e^{\beta_i \alpha_i (1 - r_i/N)}} + \mu_{i0} \right],$$

with no asymptotic approximation. Several structural consequences follow immediately: a player-specific cooperation threshold at  $r_i = N$  (under symmetric mutation), payoff-neutrality under zero choice intensity, and the sign of each player's sensitivity to their own parameters.

## 1 Introduction

The public goods game is a canonical model of cooperation in evolutionary game theory: players decide whether to contribute to a common resource, which is multiplied and redistributed equally, creating a tension between individual incentive and collective benefit [5]. Classical analyses assume a homogeneous population, but real groups are heterogeneous: players differ in their productivity, their contributions, and how sensitively they respond to payoff differences [4].

A natural learning rule for such settings is *introspection dynamics*, introduced in [1]: at each time step a player compares their current payoff to the payoff they *would have received* had they played the alternative action (keeping all other players fixed), and switches with a probability governed by this difference. Because the comparison is self-referential rather than imitative, introspection applies to arbitrary asymmetric games where role-model imitation is unavailable or unnatural.

In [2] the authors extended introspection dynamics to multiplayer asymmetric games and derived a formula for the stationary distribution that can be evaluated numerically for any game. In their general framework the transition rate for player  $i$  switching depends on the actions of all other players through the payoff function, so the stationary distribution must be computed by solving a linear system of dimension  $2^N$ ; no closed form is available in general.

In this paper we show that whenever the payoff difference  $\Delta f_i$  a player evaluates when considering a switch is independent of all other players' current actions, a property we call *state-independence*, the introspection chain decomposes exactly as a product of  $N$  independent two-state chains (one per player), the stationary distribution is a product measure, and the cooperation probability  $p_C^{\text{intro}}$  admits an explicit closed form scaling as  $O(N)$  rather than  $O(2^N)$ . We then show that the heterogeneous public goods game (PGG) satisfies state-independence as a consequence of the *linearity* of its payoff structure, yielding the exact formula (8).

The paper is organised as follows. Section 2 introduces the general framework and the state-independence condition. Section 3 proves, for any state-independent game, that individual cooperation probabilities are given by (4) and the stationary distribution is a product measure. Section 4 establishes state-independence for the PGG and derives the exact cooperation formula. Section 5 records several structural corollaries. Section 6 concludes.

## 2 Setup and notation

Consider  $N$  ordered players, each choosing from two actions  $\{C, D\}$  coded as  $\{1, 0\}$ . The state space is  $S = \{0, 1\}^N$  with  $|S| = 2^N$ . Each player  $i$  has a payoff function  $f_i : S \rightarrow \mathbb{R}$  and player-specific choice intensity  $\beta_i > 0$ . Writing  $\mathbf{a}_{i \rightarrow \bar{a}_i}$  for the state with player  $i$ 's action flipped to  $\bar{a}_i = 1 - a_i$ , the *payoff difference* for player  $i$  at state  $\mathbf{a}$  is

$$\Delta f_i(\mathbf{a}) := f_i(\mathbf{a}) - f_i(\mathbf{a}_{i \rightarrow \bar{a}_i}).$$

A payoff function  $f_i$  is *state-independent* if  $\Delta f_i(\mathbf{a})$  depends only on  $a_i$ , not on  $\mathbf{a}_{-i}$ , the actions of all individuals other than  $i$ . When this holds there is a constant  $\delta_i \in \mathbb{R}$  such that

$$\Delta f_i(\mathbf{a}) = (1 - 2a_i) \delta_i, \tag{1}$$

where  $\delta_i = \Delta f_i|_{a_i=0}$  is the payoff difference when player  $i$  defects. When  $a_i = 0$ , (1) follows directly from the definition of  $\delta_i$ ; when  $a_i = 1$ ,  $\Delta f_i|_{a_i=1} = -\Delta f_i|_{a_i=0} = -\delta_i$ , giving (1).

Note that not all games lead to *state-independent* payoff functions. For example in [1] two of the 2 player, 2 strategy games considered to illustrate introspection dynamics are: the Prisoner's Dilemma (2 -  $M_1$ ) and the Stag-hunt (2 -  $M_2$ ) for  $b > c_1 \leq c_2 > 0$ .

$$M_1 = \begin{pmatrix} b - c_1, b - c_2 & -c_1, b \\ b, -c_2 & 0, 0 \end{pmatrix} \quad M_2 = \begin{pmatrix} b - c_1, b - c_2 & -c_1, 0 \\ 0, -c_2 & 0, 0 \end{pmatrix} \tag{2}$$

For  $M_1$ :  $\delta_1 = c_1$  and  $\delta_2 = c_2$  however for  $M_2$ ,  $\Delta f_1((1, 1)) = b - c_1$  but  $\Delta f_1((1, 0)) = -c_1$ . The Prisoner's Dilemma (as defined by  $M_1$ ) is *state-independent* but the Stag-hunt game (as defined by  $M_2$ ) is not. Further games considered in [1] are the Volunteer's dilemma and the Volunteer's timing dilemma (a generalisation with more than 2 strategies) which are both not *state-independent*.

To define introspection dynamics we introduce the following two quantities:

- **Fermi function.** Each player  $i$  has a personal Fermi function  $\phi_i(x) = (1 + e^{\beta_i x})^{-1}$ . Since  $\beta_i > 0$ , the exponential  $e^{\beta_i x}$  is strictly increasing in  $x$ , so  $\phi_i$  is strictly decreasing with  $\phi_i(0) = \frac{1}{2}$  and  $\phi_i(x) + \phi_i(-x) = 1$ .
- **Mutation.** Each player  $i$  has mutation probabilities  $\mu_{i0}, \mu_{i1} > 0$  with  $\mu_{i0} + \mu_{i1} < 1$ , where  $\mu_{i0}$  is the probability of mutating to cooperation and  $\mu_{i1}$  is the probability of mutating to defection, regardless of the intended choice of the dynamic.

**Introspection dynamics.** At each step, one player  $i$  is selected uniformly at random. Player  $i$  considers switching from action  $a_i$  to the alternative  $\bar{a}_i = 1 - a_i$ . The transition probability for the switch is

$$T_{\mathbf{a}, \mathbf{a}_{i \rightarrow \bar{a}_i}}^{\text{intro}} = \frac{1}{N} [(1 - \mu_{i0} - \mu_{i1})\phi_i(\Delta f_i(\mathbf{a})) + \mu_{i, \bar{a}_i}], \quad (3)$$

where  $\mu_{i, \bar{a}_i}$  denotes  $\mu_{i0}$  when  $\bar{a}_i = C$  and  $\mu_{i1}$  when  $\bar{a}_i = D$ . Since  $\phi_i$  is strictly decreasing, a larger payoff gain from keeping the current action yields a smaller switching probability.

**Definition 1** (Cooperation probability). *Given an ergodic chain on  $S$  with stationary distribution  $\pi$ , the cooperation probability is  $p_C = \pi \cdot s$ , where  $s_{\mathbf{a}} = \frac{1}{N} \sum_i a_i$  is the fraction of cooperators in state  $\mathbf{a}$ .*

### 3 Individual cooperation probabilities

For a state-independent game,  $\Delta f_i(\mathbf{a})$  depends only on  $a_i$ , not on  $\mathbf{a}_{-i}$ . We use this to compute each player's long-run cooperation probability and the stationary distribution directly from the two-state update rule.

**Theorem 2** (Individual cooperation probabilities). *For any state-independent game with constants  $\delta_i$ , under introspection dynamics with player-specific choice intensity  $\beta_i$  and mutation probabilities  $\mu_{i0}, \mu_{i1} > 0$  with  $\mu_{i0} + \mu_{i1} < 1$ , the long-run probability that player  $i$  cooperates is*

$$p_i = \phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0}. \quad (4)$$

*Proof.* By state-independence,  $\Delta f_i(\mathbf{a}) = (1 - 2a_i)\delta_i$  depends only on  $a_i$ . When player  $i$  is selected, the probability their next action is  $C$  equals  $p_i$  regardless of their current action: if  $a_i = D$ , the switch probability is  $p_i = \phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0}$ ; if  $a_i = C$ , the probability of remaining at  $C$  is  $1 - [\phi_i(-\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i1}]$ , using  $\phi_i(x) + \phi_i(-x) = 1$  this gives:  $1 - [(1 - \phi_i(\delta_i))(1 - \mu_{i0} - \mu_{i1}) + \mu_{i1}] = p_i$ . Since each selection of player  $i$  places them in state  $C$  with probability  $p_i$  independently of their current state, the long-run cooperation probability of player  $i$  is  $p_i$ .  $\square$

**Theorem 3** (Stationary distribution). *For any state-independent game with constants  $\delta_i$ , the stationary distribution of the introspection chain on  $S = \{0, 1\}^N$  is the product measure*

$$\pi_{\mathbf{a}} = \prod_{i=1}^N p_i^{a_i} (1 - p_i)^{1-a_i}, \quad \mathbf{a} \in S, \quad (5)$$

where  $p_i$  is given by (4).

*Proof.* By Theorem 2, each selection of player  $i$  places them in state  $C$  with probability  $p_i$  independently of every other player's action. In stationarity the players' actions are therefore independent, with player  $i$  cooperating with probability  $p_i$  and defecting with probability  $1 - p_i$ . The probability of state  $\mathbf{a}$  is the product of these marginals, giving (5).  $\square$

## 4 State-independence of the heterogeneous public goods game

The *heterogeneous public goods game* is played by  $N$  players with player-specific contributions  $\alpha_1, \dots, \alpha_N > 0$  and public goods multipliers  $r_1, \dots, r_N > 1$ . The payoff to player  $i$  in state  $\mathbf{a} = (a_1, \dots, a_N)$  is

$$f_i(\mathbf{a}) = \frac{r_i}{N} \sum_{j=1}^N \alpha_j a_j - \alpha_i a_i. \quad (6)$$

We write  $C(\mathbf{a}) = \sum_j \alpha_j a_j$  for the total contribution.

**Lemma 4** (State-independence of the PGG). *The payoff (6) is state-independent for every player  $i$ , with*

$$\delta_i = \alpha_i \left(1 - \frac{r_i}{N}\right), \quad (7)$$

i.e.  $\Delta f_i(\mathbf{a}) = (1 - 2a_i) \alpha_i (1 - r_i/N)$ , independent of  $\mathbf{a}_{-i}$ .

*Proof.* When player  $i$  switches from  $a_i$  to  $\bar{a}_i = 1 - a_i$ , only their own contribution changes, so the total contribution becomes  $C(\mathbf{a}_{i \rightarrow \bar{a}_i}) = C(\mathbf{a}) + \alpha_i(1 - 2a_i)$ . Expanding both payoffs:

$$\begin{aligned} f_i(\mathbf{a}) &= \frac{r_i}{N} C(\mathbf{a}) - \alpha_i a_i, \\ f_i(\mathbf{a}_{i \rightarrow \bar{a}_i}) &= \frac{r_i}{N} [C(\mathbf{a}) + \alpha_i(1 - 2a_i)] - \alpha_i(1 - a_i). \end{aligned}$$

Subtracting:

$$\begin{aligned} \Delta f_i &= -\frac{r_i}{N} \alpha_i (1 - 2a_i) + \alpha_i (1 - 2a_i) \\ &= \alpha_i (1 - 2a_i) \left(1 - \frac{r_i}{N}\right). \end{aligned}$$

The term  $C(\mathbf{a})$  cancels entirely, confirming state-independence with  $\delta_i = \alpha_i(1 - r_i/N)$ . □

**Corollary 5** (Exact cooperation probability). *For the heterogeneous public goods game (6), the long-run cooperation probability is*

$$p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N \left[ \frac{1 - \mu_{i0} - \mu_{i1}}{1 + e^{\beta_i \alpha_i (1 - r_i/N)}} + \mu_{i0} \right]. \quad (8)$$

*Proof.* By Lemma 4,  $\delta_i = \alpha_i(1 - r_i/N)$ . By Theorem 2:  $p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N p_i$ . Substituting  $\phi_i(\delta_i) = (1 + e^{\beta_i \alpha_i (1 - r_i/N)})^{-1}$  into (4) gives (8). □

Figure 1 validates Corollary 5 against exact values and simulation across group sizes. Table 1 verifies Theorem 3: for  $N = 3$  with per-player multipliers  $r_1 = 1$ ,  $r_2 = 3 = N$ ,  $r_3 = 9$ , uniform contributions  $\alpha_i = 1$ ,  $\beta = 2$ , and  $\mu_{i0} = \mu_{i1} = 0.1$ , the individual cooperation probabilities are  $p_1 \approx 0.267$ ,  $p_2 = 0.500$ ,  $p_3 \approx 0.886$ , and the formula and exact values of  $\pi_{\mathbf{a}}$  agree to five decimal places for all eight states.

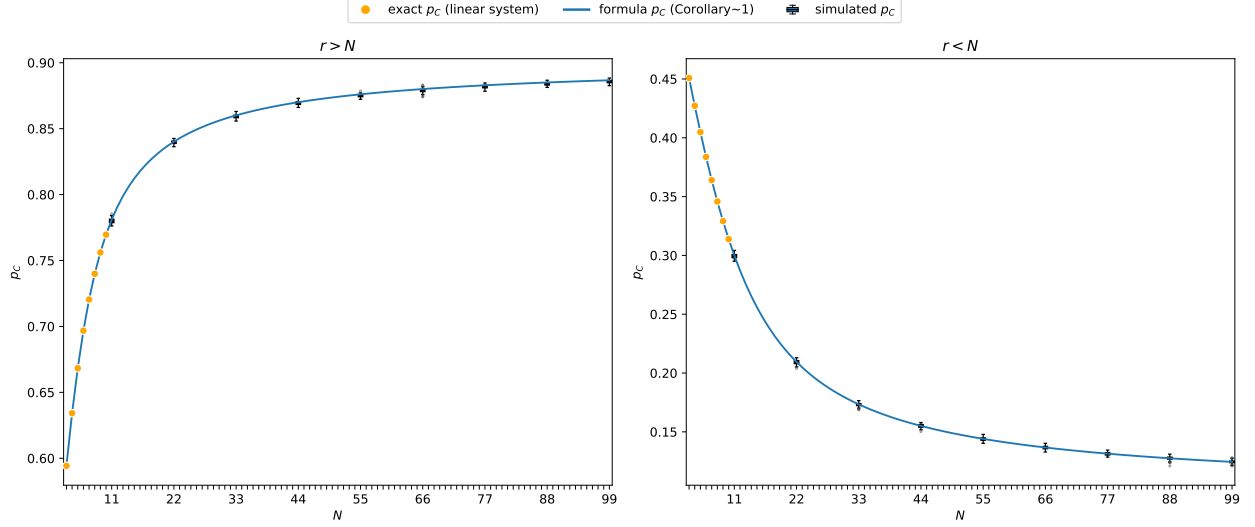


Figure 1: Validation of formula (8) on a heterogeneous group with  $\alpha_i = i$ ,  $\beta = 0.5$ ,  $\mu_{i0} = \mu_{i1} = 0.1$ . *Left:*  $r = 2N$  (so  $r > N$  for all  $N$ ). *Right:*  $r = N/2$  (so  $r < N$  for all  $N$ ). Orange dots: exact  $p_C$  from the full  $2^N$  linear system (feasible only for small  $N$ ). Blue curve: formula (8) (Corollary 5). Blue boxes: distribution of  $p_C$  across 99 independent runs of  $10^5$  simulation steps each (outliers shown). The negligible variability across runs reflects the large number of simulation steps per run. All values obtained using [3].

## 5 Structural consequences

The formulas (4) and (8) allow immediate structural conclusions about individual and aggregate cooperation.

**Corollary 6** (Player-specific cooperation thresholds). *Player  $i$  cooperates with probability greater than  $\frac{1}{2}$  if and only if*

$$\phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

*For the PGG,  $\delta_i = \alpha_i(1 - r_i/N)$ ; when  $\mu_{i0} = \mu_{i1}$  this reduces to  $r_i > N$ . A sufficient condition for  $p_C^{\text{intro}} > \frac{1}{2}$  is that the inequality holds for every  $i$ .*

*Proof.* Since  $1 - \mu_{i0} - \mu_{i1} > 0$ :

$$p_i > \frac{1}{2} \iff (1 - \mu_{i0} - \mu_{i1})\phi_i(\delta_i) > \frac{1}{2} - \mu_{i0} \iff \phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

When  $\mu_{i0} = \mu_{i1}$ , the right-hand side equals  $\frac{1}{2}$ , and since  $\phi_i$  is strictly decreasing with  $\phi_i(0) = \frac{1}{2}$ , this is equivalent to  $\delta_i < 0$ . For the PGG,  $\delta_i = \alpha_i(1 - r_i/N) < 0$  iff  $r_i > N$ . The claim about  $p_C^{\text{intro}}$  follows by averaging. □

**Corollary 7** (Neutral drift). *If  $\beta_i = 0$  for all  $i$ , then  $\phi_i$  is constant and the payoff parameters  $\delta_i$  have no effect on  $p_C$ ; cooperation is determined entirely by mutation:*

$$p_i = \frac{1 + \mu_{i0} - \mu_{i1}}{2}, \quad p_C^{\text{intro}} = \frac{1}{2} + \frac{1}{2N} \sum_{i=1}^N (\mu_{i0} - \mu_{i1}).$$

| State $\mathbf{a}$ | Formula $\pi_{\mathbf{a}}$ | Exact $\pi_{\mathbf{a}}$ |
|--------------------|----------------------------|--------------------------|
| DDD                | 0.04193                    | 0.04193                  |
| DDC                | 0.32463                    | 0.32462                  |
| DCD                | 0.04193                    | 0.04193                  |
| DCC                | 0.32463                    | 0.32462                  |
| CDD                | 0.01526                    | 0.01527                  |
| CDC                | 0.11818                    | 0.11818                  |
| CCD                | 0.01526                    | 0.01527                  |
| CCC                | 0.11818                    | 0.11818                  |

Table 1: Stationary-distribution probabilities  $\pi_{\mathbf{a}}$  for all eight states  $\mathbf{a} \in \{D, C\}^3$  with  $N = 3$ ,  $\alpha_i = 1$ ,  $\beta = 2$ ,  $\mu_{i0} = \mu_{i1} = 0.1$ , and per-player multipliers  $r_1 = 1 < N = r_2 = 3 < r_3 = 9$ . Formula: product measure (5) (Theorem 3); Exact: values from the  $2^N$  linear system obtained using [3]. Both columns to five decimal places.

In particular,  $p_C^{\text{intro}} = \frac{1}{2}$  if and only if  $\sum_i (\mu_{i0} - \mu_{i1}) = 0$ .

*Proof.* When  $\beta_i = 0$ , the Fermi function satisfies  $\phi_i(x) = \frac{1}{2}$  for all  $x$ . Substituting into (4):

$$p_i = \frac{1 - \mu_{i0} - \mu_{i1}}{2} + \mu_{i0} = \frac{1}{2} + \frac{\mu_{i0} - \mu_{i1}}{2} = \frac{1 + \mu_{i0} - \mu_{i1}}{2}. \quad \square$$

**Corollary 8** (Role of heterogeneity). *For the PGG, for player  $i$  with  $r_i \neq N$ :*

- (i)  $\frac{\partial p_i}{\partial \alpha_i}$  is positive when  $r_i > N$  and negative when  $r_i < N$ .
- (ii)  $\frac{\partial p_i}{\partial \beta_i}$  is positive when  $r_i > N$  and negative when  $r_i < N$ .

*Proof.* Write  $p_i = (1 - \mu_{i0} - \mu_{i1})\phi_i(\alpha_i(1 - r_i/N)) + \mu_{i0}$ .

Part (i). Differentiating with respect to  $\alpha_i$ :

$$\frac{\partial p_i}{\partial \alpha_i} = (1 - \mu_{i0} - \mu_{i1})(1 - r_i/N) \phi'_i(\alpha_i(1 - r_i/N)).$$

Since  $\phi'_i(x) < 0$  for all  $x$  and  $1 - \mu_{i0} - \mu_{i1} > 0$ , the sign equals  $-\text{sgn}(1 - r_i/N) = \text{sgn}(r_i/N - 1)$ , which is positive if and only if  $r_i > N$ .

Part (ii). Write  $g(t) = (1 + e^t)^{-1}$ , so  $\phi_i(x) = g(\beta_i x)$ . Differentiating with respect to  $\beta_i$ :

$$\frac{\partial p_i}{\partial \beta_i} = (1 - \mu_{i0} - \mu_{i1}) \alpha_i(1 - r_i/N) g'(\beta_i \alpha_i(1 - r_i/N)).$$

Since  $g' < 0$  and  $\alpha_i > 0$ , the sign equals  $-\text{sgn}(1 - r_i/N) = \text{sgn}(r_i/N - 1)$ , which is positive if and only if  $r_i > N$ .  $\square$

**Remark 9.** Corollaries 6–8 show that each player's long-run cooperation is fully determined by their individual triple  $(\alpha_i, \beta_i, r_i)$  and the group size  $N$ . Players with  $r_i > N$  cooperate more than half the time, and increasing  $\alpha_i$  or  $\beta_i$  raises their cooperation probability further; players with  $r_i < N$  show the opposite sensitivity. The parameters of other players have no effect on player  $i$ 's marginal cooperation probability.

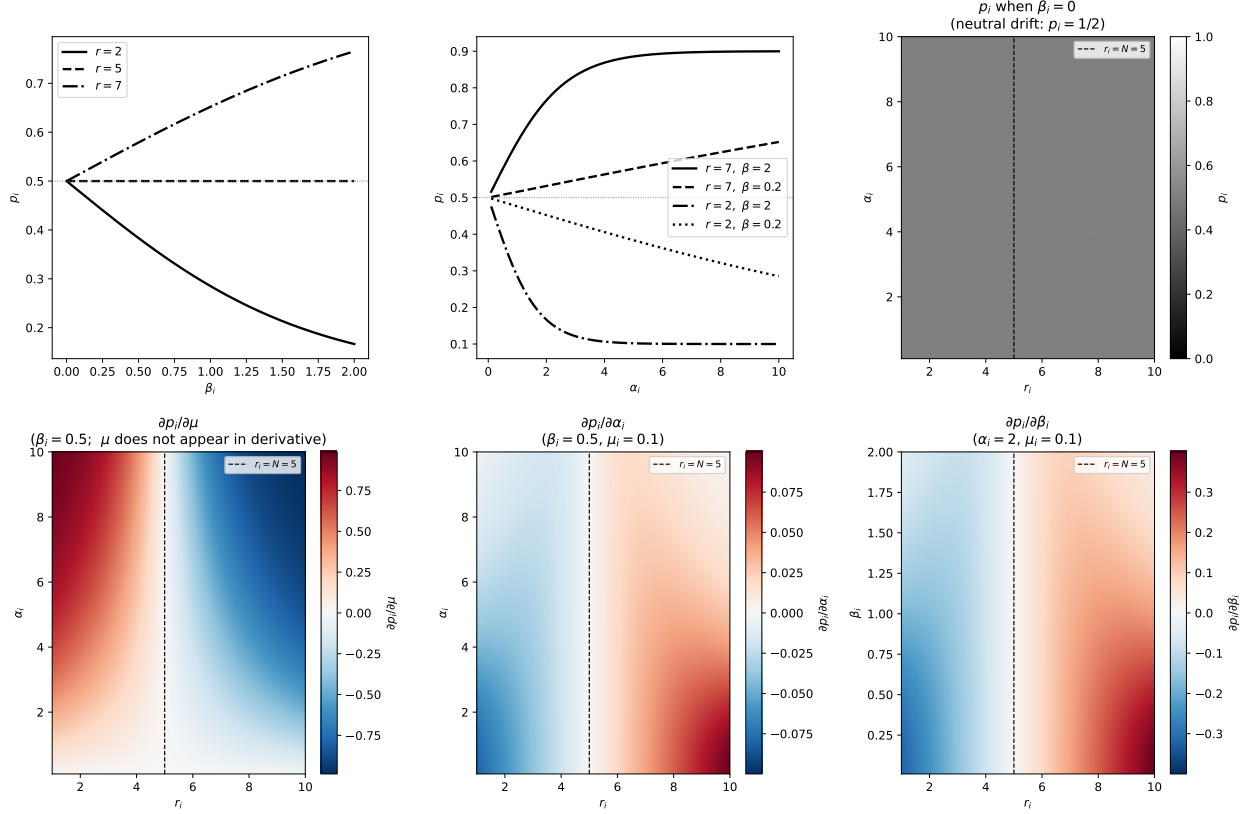


Figure 2: Structural properties of  $p_i$  for a single player with  $N = 5$ ,  $\mu_{i0} = \mu_{i1} = 0.1$ . *Top left:*  $p_i$  as a function of  $\beta_i$  ( $\alpha_i = 2$ ) for  $r < N$ ,  $r = N$ , and  $r > N$ ; all curves pass through  $p_i = \frac{1}{2}$  at  $\beta_i = 0$  (Corollary 7). *Top centre:*  $p_i$  as a function of  $\alpha_i$  for two values of  $r$  and  $\beta_i$ ; curves above (below)  $\frac{1}{2}$  correspond to  $r > N$  ( $r < N$ ). *Top right:*  $p_i(\alpha_i, r_i)$  at  $\beta_i = 0$ , confirming  $p_i = \frac{1}{2}$  everywhere (Corollary 7); the dashed line marks  $r_i = N$ . *Bottom row:* Heatmaps of the partial derivatives  $\partial p_i / \partial \mu_i$ ,  $\partial p_i / \partial \alpha_i$ , and  $\partial p_i / \partial \beta_i$  over the  $(r_i, \alpha_i)$  or  $(r_i, \beta_i)$  plane ( $\beta_i = 0.5$  or  $\alpha_i = 2$  as fixed parameter). Red indicates a positive derivative; blue indicates negative. Each derivative changes sign exactly at  $r_i = N$  (dashed line), consistent with Corollary 8.

## 6 Conclusion

We have proved that introspection dynamics on any state-independent game is exactly solvable. When  $\Delta f_i$  depends only on  $a_i$ , each selection of player  $i$  places them in state  $C$  with probability  $p_i$  regardless of current state (Theorem 2), and the stationary distribution is a product measure (Theorem 3). For the heterogeneous public goods game, the linear payoff structure implies state-independence (Lemma 4), and the long-run cooperation probability has a closed form with  $N$  terms rather than  $2^N$  (Corollary 5).

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